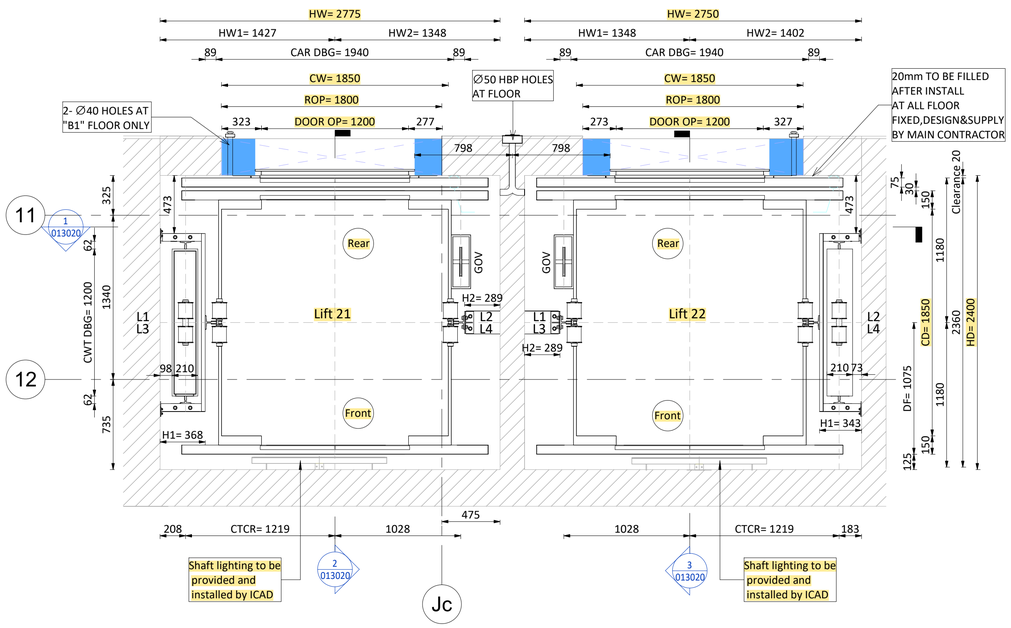
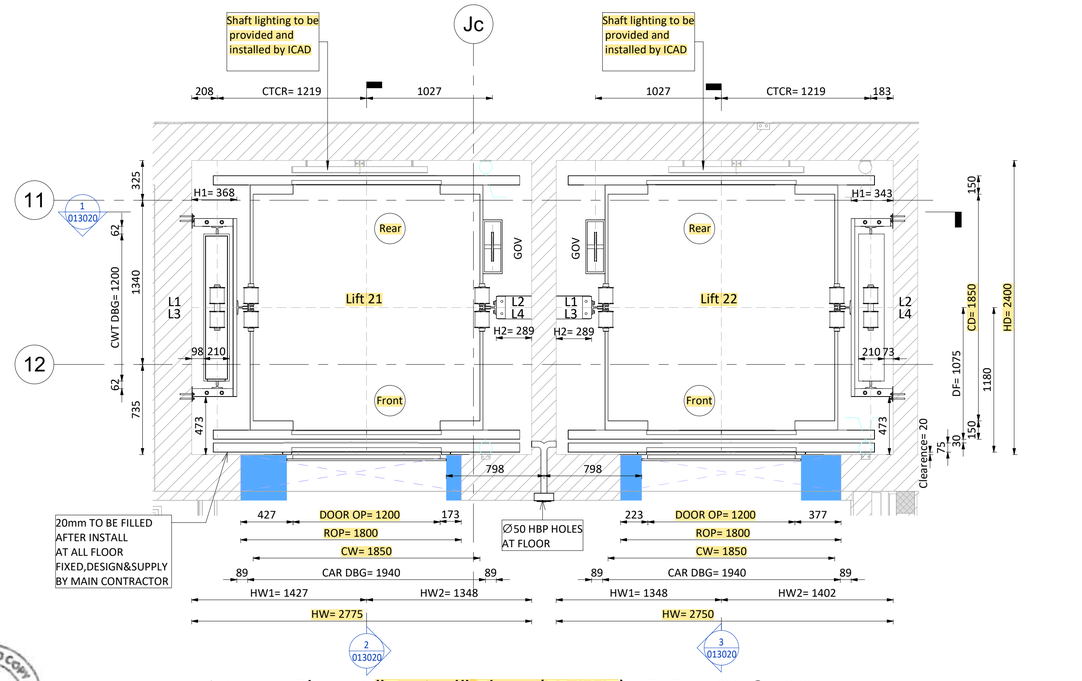
Lift 21,22



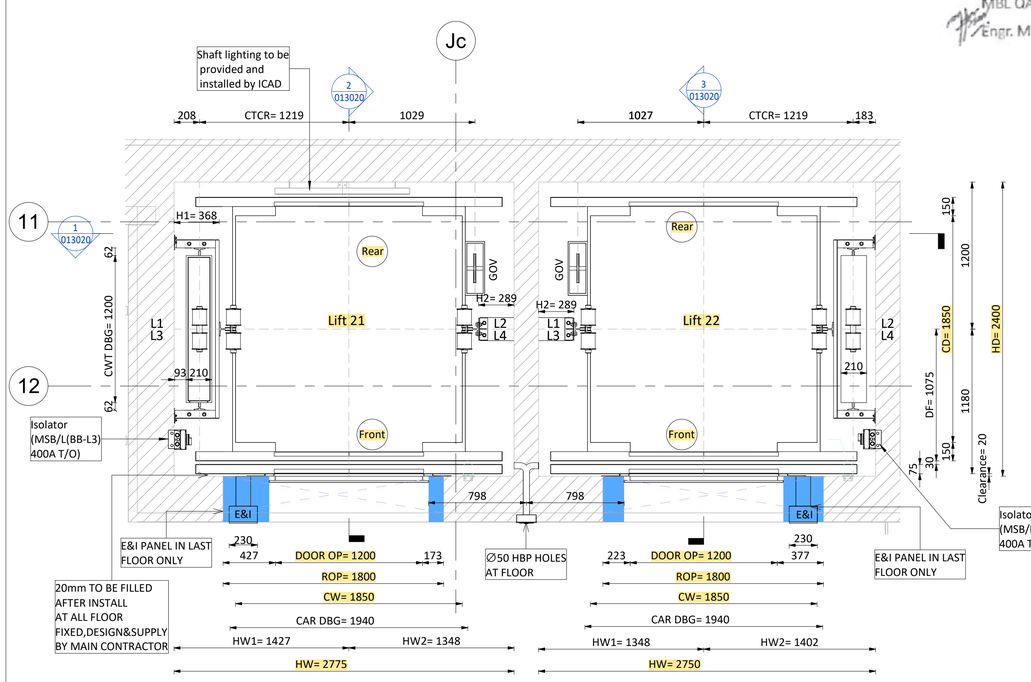
1 Hoist Way Plan At "B1" Floor (REAR) - ELEV-L21 & L22

1 : 25



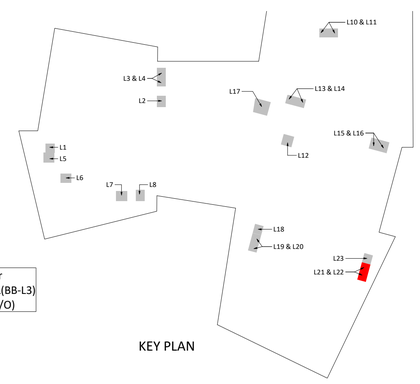
2 Hoist Way Plan At "Typical" Floor (FRONT) - ELEV-L21 & L22

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3 Hoist Way Plan At "3" Floor (FRONT) - ELEV-L21 & L22

1 : 25

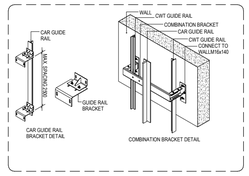


MBL to provide a comprehensive Method Statement describing the procedure for filling all gaps and correcting / adjusting the rough openings for doors

dar Dar Al-Handasah consultants (Shair and Partners)
Checked and Reviewed

SCEC – K&A Reviewed
Civil | Arch. | Elec. | ELV. | Mech. | Acoustic

SCEC – K&A Reviewed
Code B – Approved with Comments
(Do not Resubmit)



ELEV-L21 & L22 ELEVATOR GENERAL SPECIFICATIONS

LIFT NUMBER	ELEV-L21 & L22
CABIN INTERNAL SIZE(WXDXH)	1850X1850X2600
CAPACITY	1600 Kg
CLEAR ENTRANCE	1200X2400
CONTROL	VVVF
CONTROL SYSTEM	DUPLEX FULL COLLECTIVE
COUNTERWEIGHT SAFETY	NO
DOOR TYPE	CLD1
EMERGENCY EXITS	-
FRONT OPENING	4
HEAT RELEASE	1616Kcal/h
HOISTWAY SIZE(HWXHD)	2775X2400(L21)& 2750X2400(L22)
LIFT NUMBER	ELEV-L21 & L22
LIGHTING SUPPLY	220V 60HZ
MIN.FLOOR HEIGHT	-
NO. OF FLOOR	5
NO. OF OPENINGS	5
NO. OF UNITS	2
NON-STOP FLOORS	-
NON-STOP FLOORS NAMES	-
NUMBER OF PASSENGERS	21P
OVERHEAD	5400mm
PIT DEPTH	1600mm
POWER SUPPLY	380V 3P 60HZ
PRODUCT NO.	GeN2 MRL
RATED CURRENT(A)	25.6
REAR OPENING	1
ROPING	2:1
SPEED(m/s)	1.00
START CURRENT(A)	28.4
SYSTEM POWER(KW)	13.5
TRAVEL	18025mm
USE	PASSENGER LIFT Public

REACTIONS FORCE (KN)

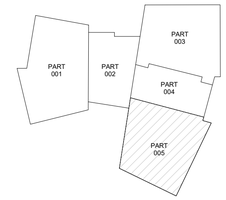
Car side rail	CWT guide rail	Pit reaction
R= 41	R= 15	U= 165
R1= 2.47	R1= 1.13	V= 132
R2= 3.14	R2= 0.3	

LEGEND

HW	HOIST WAY WIDTH
HD	HOIST WAY DEPTH
CRH	CAR ROUGH HEIGHT
DOOR OP	DOOR OPENING
ROP	ROUGH OPENING
CW	CAR WIDTH
CD	CAR DEPTH
CH	CAR HEIGHT
DBG	DISTANCE BETWEEN GUARDRAIL
CWT	COUNTERWEIGHT
E&I PANEL	EMERGENCY & INSPECTION PANEL
G.O.V	CAR GOVERNOR
COP	CAR OPERATING PANEL
HBP	HALL POSITION INDICATOR

NOTE No.:
FRONT WALLS,FINISHES & ARCHITRAVES TO BE BUILT (BY CLIENT) AFTER THE INSTALLATION OF THE (LANDING) EXTERNAL DOORS.

IMPORTANT NOTE FOR MAIN CONTRACTOR.
FREEDOM OF ACCESS MUST BE AVAILABLE FOR POINT OF ENTRY INTO THE BUILDING UP TO THE ELEVATOR HOIST WAY AND THE FRONT WALLS OF THE HOIST WAY AT MAIN FLOOR TO BE LEFT OPEN FOR FULL WIDTH AND HEIGHT TO PERMIT ENTRY OF OUR EQUIPMENT.



KEY PLAN

GENERAL NOTES: ALL LEVELS SHOWN ON THE DRAWING ARE CHART DATUM LEVELS PROJECT LEVEL+0.000 = +3.700 CHART DATUM.

Refer to :
LIFTS overhead steel hook beams structural calculation and fixation details: TR-OPE-MBL-JCDC-05193-S01
CRS: SD-OPE-MBL-JCDC-01334_CRS- KA comment_1-version-(0_00)-L21,22
Electrical Panels distribution: JOH-ICD-S5-01-056-XXX-03-SD-EL-011630_1
wiring diagram opera lift
Acoustic Letter: L-OTIS-MBL-OPE-26-0015

DRAWING / DOCUMENT REFERENCE

NO.	NUMBER	TITLE	REV.
01	L-21 & L-22	JOH-HLA-S4-01-056-005-00-DR-AR-011105	F00
02		JOH-BHC-S4-01-056-XXX-XX-DR-VT-015001	
03		JOH-BHC-S4-01-056-XXX-XX-DR-VT-010001	
04		JOH-BHC-S4-01-056-005-ZZ-DR-VT-013001	
05		JOH-BHC-S4-01-056-005-ZZ-DR-VT-011001	
06		TQ-OPE-MBL-JCDC-00222	
07		JCD-OPE-MBL-MR-003-NMR-601-03	
08		JCD-OPE-MBL-MR-003-NMR-602-01	
09		TR-OPE-MBL-JCDC-05193-01	

REVIEWED BY DISCIPLINE	SIGNATURE	DATE
ARCHITECTURAL	MT	[illegible]
STRUCTURAL	PC	[illegible]
MECHANICAL	Sh.A	2026-05-20
PLUMBING	Sh.A	2026-05-20
ELECTRICAL	Sh.A	[illegible]
ICT	SH.A	2026-05-[illegible]
ADDITIONAL DISCIPLINES TO BE ADDED AS REQUIRED		

REV	DATE	BY	DESCRIPTION	CHK	APD
S01	05-MAY-2026	AK	Shop Drawing	SS	AB
S00	05-MAY-2026	AK	Issued for Approval	SS	AB

DRAWING STATUS: CONSTRUCTION SHOPDRAWING

CLIENT: شركة جدة للتطوير Jeddah Central Development Co.

LEAD DESIGN CONSULTANTS: Henning Larsen

LEAD ENGINEERING CONSULTANT: BURO HAPPOLD

PROJECT SUPERVISION CONSULTANT: KHATIB & ALAMI

GENERAL CONTRACTOR: MBL

REVIEWER: dar

SUB CONTRACTOR: OTIS ELEVATOR COMPANY (SAUDI ARABIA) LIMITED

PROJECT: JEDDAH OPERA HOUSE

TITLE: OPERA BUILDING
VERTICAL TRANSPORTATION
PART 005
PLANS - LIFT SHAFT SHEET 01 OF 02 (L21 & L22)

SCALE @ A1:	CHECKED:	APPROVED:
1:25	SS	AB

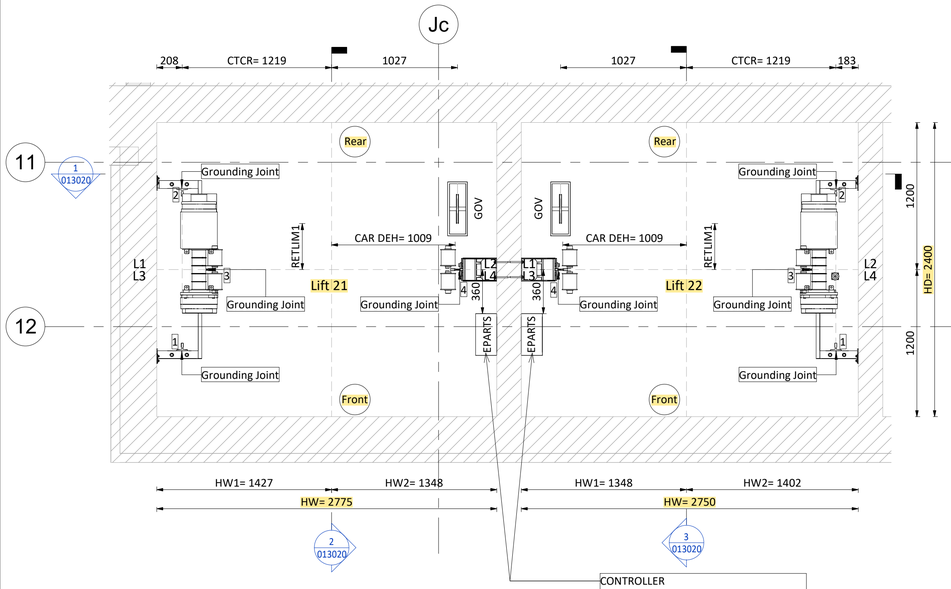
PROJECT No:	DESIGNED:	DRAWN:	DATE:
440014	AK	PR	05-MAY-2026

DRAWING No: JOH-OTS-S5-01-056-005-ZZ-SD-VT-011020 REV: S01

MODEL FILE REF: JOH-OTS-S5-01-056-XXX-ZZ-M3-VT-010001

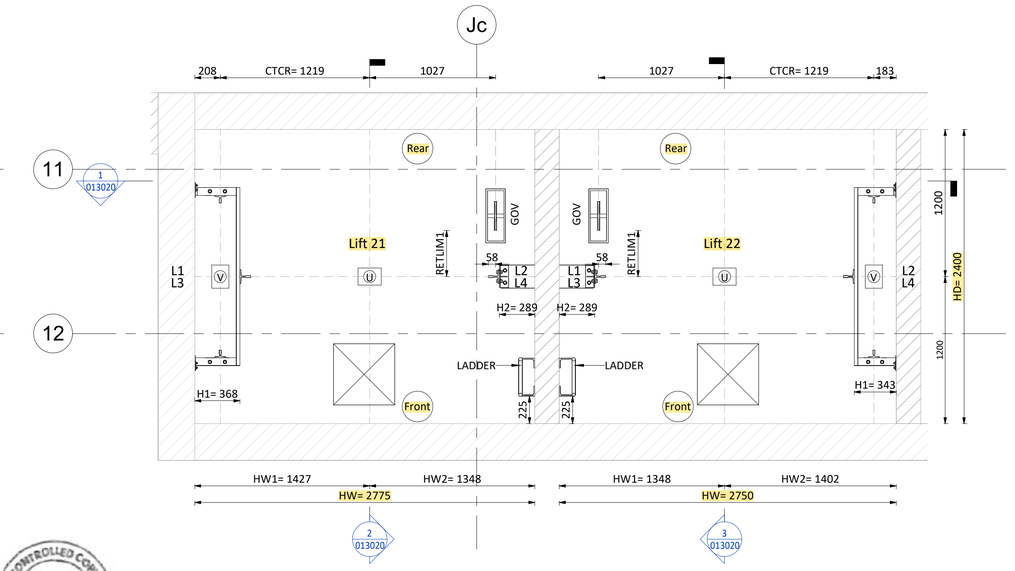
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MBL QAM
Engr. Mudassar Altaf



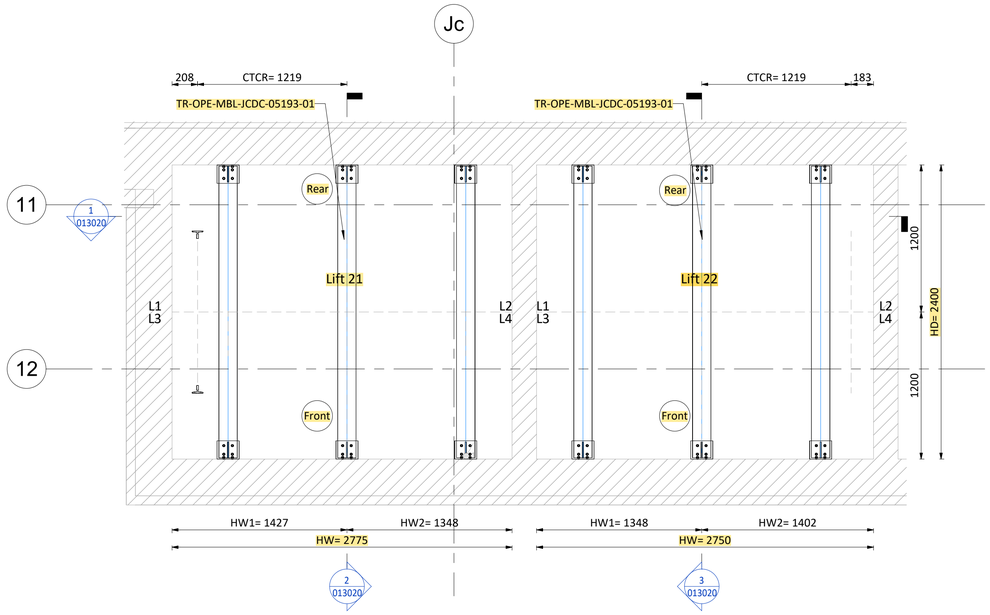
1 Machine Plan Level - ELEV-L21 & L22

1 : 25



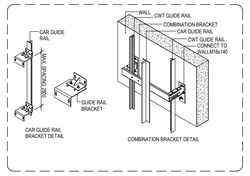
2 Hoist Way Pit Plan Floor - ELEV-L21 & L22

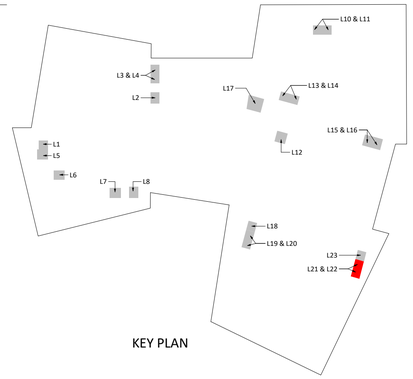
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3 Hoist Way Hook Plan - ELEV-L21 & L22

1 : 25

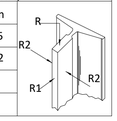




KEY PLAN

ELEV-L21 & L22 ELEVATOR GENERAL SPECIFICATIONS	
LIFT NUMBER	ELEV-L21 & L22
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CAPACITY	1600 Kg
CLEAR ENTRANCE	1200X2400
CONTROL	VVVF
CONTROL SYSTEM	DUPLEX FULL COLLECTIVE
COUNTERWEIGHT SAFETY	NO
DOOR TYPE	CLD1
EMERGENCY EXITS	-
FRONT OPENING	4
HEAT RELEASE	1616Kcal/h
HOISTWAY SIZE(HWXHD)	2775X2400(L21)& 2750X2400(L22)
LIFT NUMBER	ELEV-L21 & L22
LIGHTING SUPPLY	220V 60HZ
MIN.FLOOR HEIGHT	-
NO. OF FLOOR	5
NO. OF OPENINGS	5
NO. OF UNITS	2
NON-STOP FLOORS	-
NON-STOP FLOORS NAMES	-
NUMBER OF PASSENGERS	21P
OVERHEAD	5400mm
PIT DEPTH	1600mm
POWER SUPPLY	380V 3P 60HZ
PRODUCT NO.	GeN2 MRL
RATED CURRENT(A)	25.6
REAR OPENING	1
ROPING	2:1
SPEED(m/s)	1.00
START CURRENT(A)	28.4
SYSTEM POWER(KW)	13.5
TRAVEL	18025mm
USE	PASSENGER LIFT

REACTIONS FORCE (KN)			
Car side rail	CWT guide rail	Pit reaction	
R= 41	R= 15	U= 165	
R1= 2.47	R1= 1.13	V= 132	
R2= 3.14	R2= 0.3		



LEGEND	
HW	HOIST WAY WIDTH
HD	HOIST WAY DEPTH
CRH	CAR ROUGH HEIGHT
DOOR OP	DOOR OPENING
ROP	ROUGH OPENING
CW	CAR WIDTH
CD	CAR DEPTH
CH	CAR HEIGHT
DBG	DISTANCE BETWEEN GUARDRAIL
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E&I PANEL	EMERGENCY & INSPECTION PANEL
G.O.V	CAR GOVERNOR
COP	CAR OPERATING PANEL
HBP	HALL POSITION INDICATOR

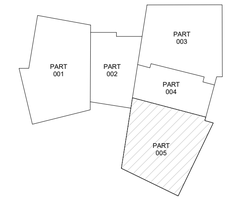
NOTE No.:
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MBL IFC
JEDDAH OPERA HOUSE PROJECT
QA/QC Department
MBL QAM
Engr. Mudassar Altaf

SCEC – K&A Reviewed
Civil | Arch. | Elec. | ELV. | Mech. | Acoustic

SCEC – K&A Reviewed
Code B – Approved with Comments
(Do not Resubmit)



KEY PLAN

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CRS: SD-OPE-MBL-JCDC-01334_CRS- KA comment_1-version-(0_00)-L21,22
Electrical Panels distribution:
JOH-ICD-S5-01-056-XXX-03-SD-EL-011630_1
wiring diagram opera lift
Acoustic Letter: L-OTIS-MBL-OPE-26-0015

DRAWING / DOCUMENT REFERENCE

NO.	NUMBER	TITLE	REV.
01	L-21 & L-22	JOH-HLA-S4-01-056-005-00-DR-AR-011105	F00
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04		JOH-BHC-S4-01-056-005-ZZ-DR-VT-013001	
05		JOH-BHC-S4-01-056-005-ZZ-DR-VT-011001	
06		TQ-OPE-MBL-JCDC-00222	
07		JCD-OPE-MBL-MR-003-NMR-601-03	
08		JCD-OPE-MBL-MR-003-NMR-602-01	
09		TR-OPE-MBL-JCDC-05193-01	

REVIEWED BY DISCIPLINE	SIGNATURE	DATE
ARCHITECTURAL	MT	[illegible]
STRUCTURAL	PC	[illegible]
MECHANICAL	SbA	2026-05-20
PLUMBING	SbA	2026-05-20
ELECTRICAL	SbA	[illegible]
ICT	SbA	[illegible]
ADDITIONAL DISCIPLINES TO BE ADDED AS REQUIRED		

REV	DATE	BY	DESCRIPTION	CHK	APD
S01	05-MAY-2026	AK	Shop Drawing	SS	AB

DRAWING STATUS: CONSTRUCTION SHOPDRAWING

CLIENT: Jeddah Central Development Co.

LEAD DESIGN CONSULTANTS: Henning Larsen

LEAD ENGINEERING CONSULTANT: BURO HAPPOLD

PROJECT SUPERVISION CONSULTANT: KHATIB & ALAMI

GENERAL CONTRACTOR: MBL

REVIEWER: dar

SUB CONTRACTOR: OTIS ELEVATOR COMPANY SAUDI ARABIA LIMITED

PROJECT: JEDDAH OPERA HOUSE

TITLE:
OPERA BUILDING
VERTICAL TRANSPORTATION
PART 005
PLANS - LIFT SHAFT SHEET 02 OF 02 (L21 & L22)

SCALE @ A1:	CHECKED:	APPROVED:
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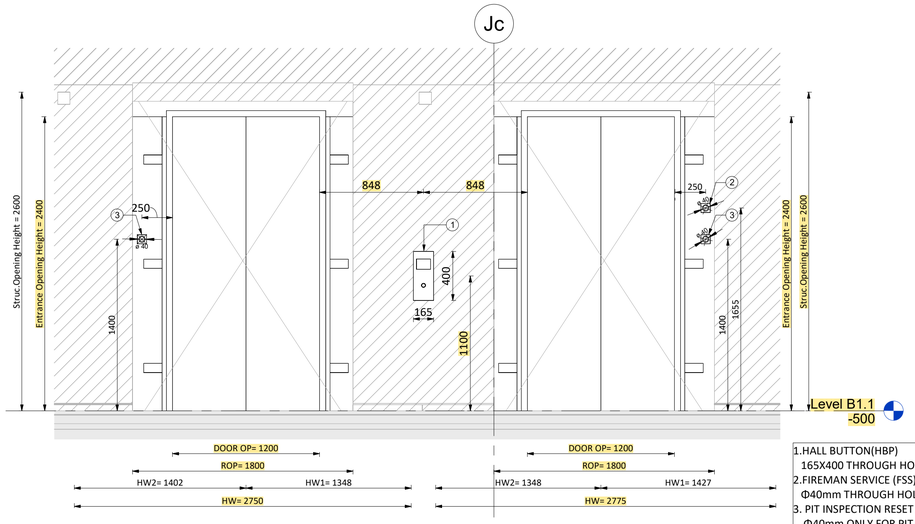
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440014	AK	PR	05-MAY-2026

DRAWING No: JOH-OTS-S5-01-056-005-ZZ-SD-VT-011021 REV: S01

MODEL FILE REF: JOH-OTS-S5-01-056-XXX-ZZ-M3-VT-010001

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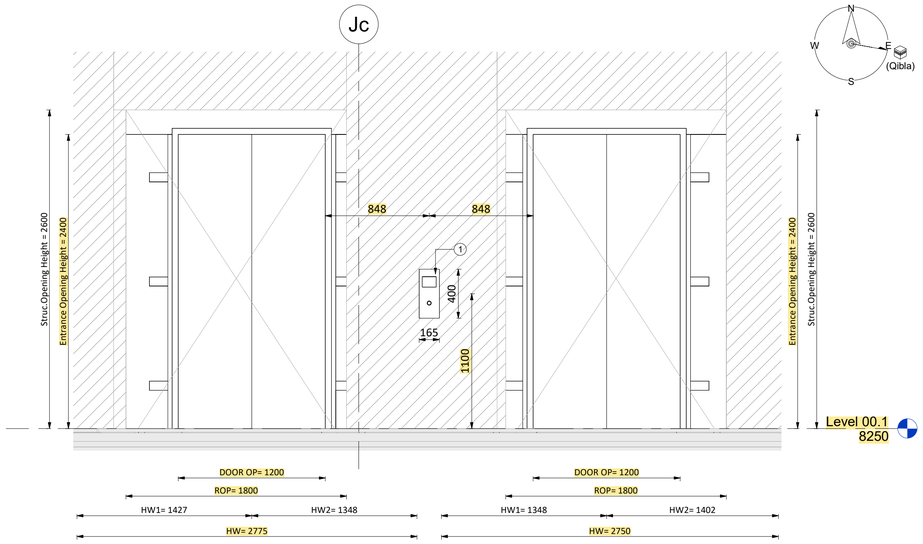
Public



1.HALL BUTTON(HBP)
165X400 THROUGH HOLE ,IN THE DISCRETIONARY FLOOR
2.FIREMAN SERVICE (FSS)
Φ40mm THROUGH HOLE, IN THE MAIN FLOOR
3. PIT INSPECTION RESET SW
Φ40mm ONLY FOR PIT LANDING
4.E&I PANEL.

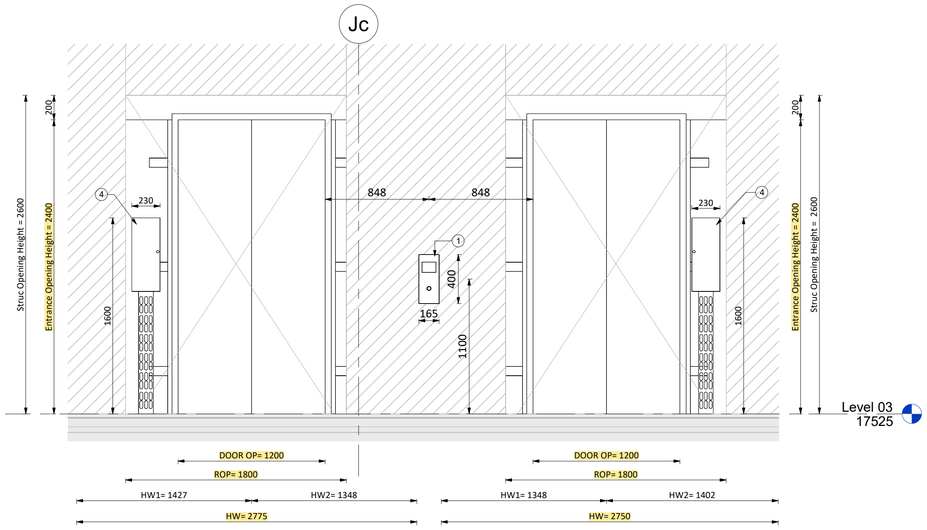
1 Door Elevation At "B1" Floor (REAR) - ELEV-L21 & L22

1 : 25



2 Door Elevation At "Typical" Floor (FRONT) - ELEV-L21 & L22

1 : 25

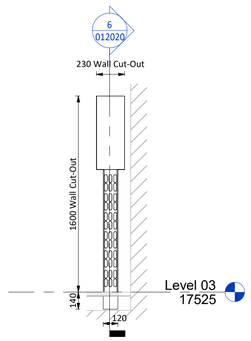


3 Door Elevation At "3" Floor (FRONT) - ELEV-L21 & L22

1 : 25

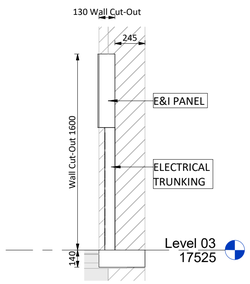
4 Cross-Section View A-A - ELEV-L21 & L22

1 : 25



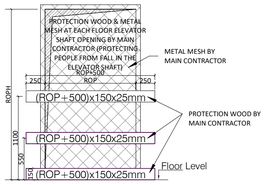
5 E&I SPOT DETAIL - ELEV-L21 & L22

1 : 25

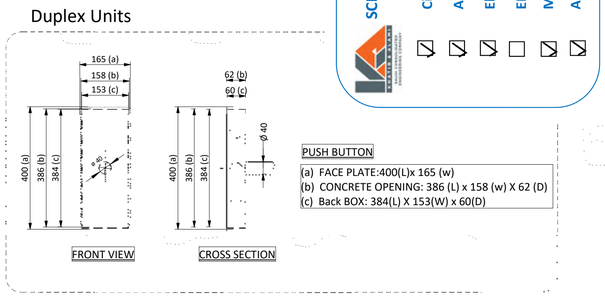


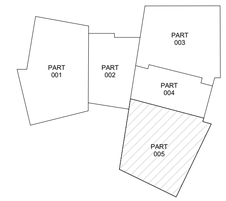
6 Cross-Section View B-B - ELEV-L21 & L22

1 : 25



Protection Elevation Detail
At Each Elevator Shaft Opening





KEY PLAN

GENERAL NOTES: ALL LEVELS SHOWN ON THE DRAWING ARE CHART DATUM LEVELS PROJECT LEVEL+0.000 = +3.700 CHART DATUM.

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LIFTS overhead steel hook beams structural calculation and fixation details: TR-OPE-MBL-JCDC-05193-S01
CRS: SD-OPE-MBL-JCDC-01334_CRS- KA comment_1-version-(0_00)-L21,22
Electrical Panels distribution:
JOH-ICD-S5-01-056-XXX-03-SD-EL-011630_1
wiring diagram opera lift
Acoustic Letter: L-OTIS-MBL-OPE-26-0015

DRAWING / DOCUMENT REFERENCE

NO.	NUMBER	TITLE	REV.
01	L-21 & L-22	JOH-HLA-S4-01-056-005-00-DR-AR-011105	F00
02		JOH-BHC-S4-01-056-XXX-XX-DR-VT-015001	
03		JOH-BHC-S4-01-056-XXX-XX-DR-VT-010001	
04		JOH-BHC-S4-01-056-005-ZZ-DR-VT-013001	
05		JOH-BHC-S4-01-056-005-ZZ-DR-VT-011001	
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REVIEWED BY DISCIPLINE	SIGNATURE	DATE
ARCHITECTURAL	MT	[illegible]
STRUCTURAL	PC	[illegible]
MECHANICAL	SbA	2026-05-20
PLUMBING	SbA	2026-05-20
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ICT	SbA	2026-05-[illegible]
ADDITIONAL DISCIPLINES TO BE ADDED AS REQUIRED		

REV	DATE	BY	DESCRIPTION	CHK	APD
S01	05-MAY-2026	AK	Shop Drawing	SS	AB
S00	05-MAY-2026	AK	Issued for Approval	SS	AB

DRAWING STATUS: CONSTRUCTION SHOPDRAWING

CLIENT: شركة وسط جدة للتطوير Jeddah Central Development Co.

LEAD DESIGN CONSULTANTS: Henning Larsen

LEAD ENGINEERING CONSULTANT: BURO HAPPOLD

PROJECT SUPERVISION CONSULTANT: KHATIB & ALAMI

GENERAL CONTRACTOR: MBL

REVIEWER: dar

SUB CONTRACTOR: OTIS ELEVATOR COMPANY SAUDI ARABIA LIMITED — OTIS

PROJECT: JEDDAH OPERA HOUSE

TITLE: OPERA BUILDING
VERTICAL TRANSPORTATION
PART 005
ELEVATIONS - LIFT SHAFT (L21 & L22)

SCALE @ A1:	CHECKED:	APPROVED:
1:25	SS	AB

PROJECT No:	DESIGNED:	DRAWN:	DATE:
440014	AK	PR	05-MAY-2026

DRAWING No: JOH-OTS-S5-01-056-005-ZZ-SD-VT-012020 REV: S01

MODEL FILE REF: JOH-OTS-S5-01-056-XXX-ZZ-M3-VT-010001

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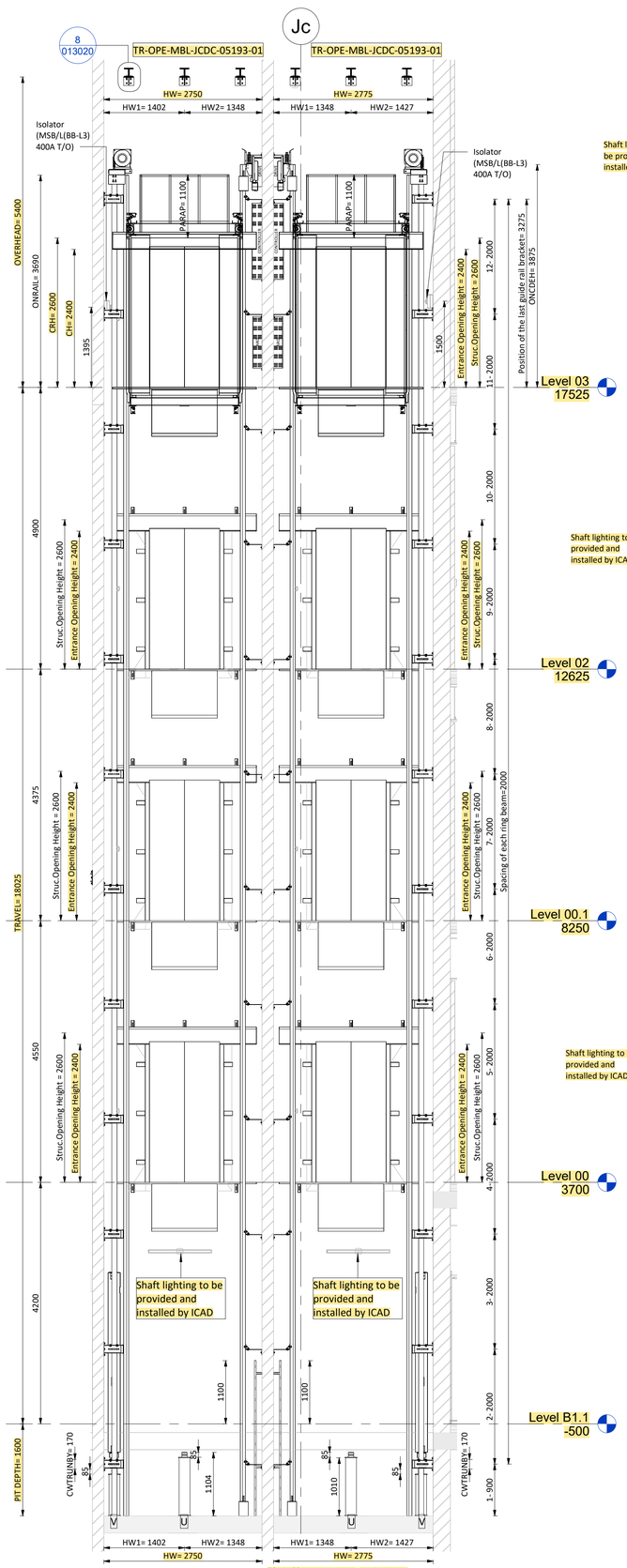
SCEC – K&A Reviewed
Civil Arch. Elec. ELV. Mech. Acoustic

SCEC – K&A Reviewed
Code B – Approved with Comments
(Do not Resubmit)

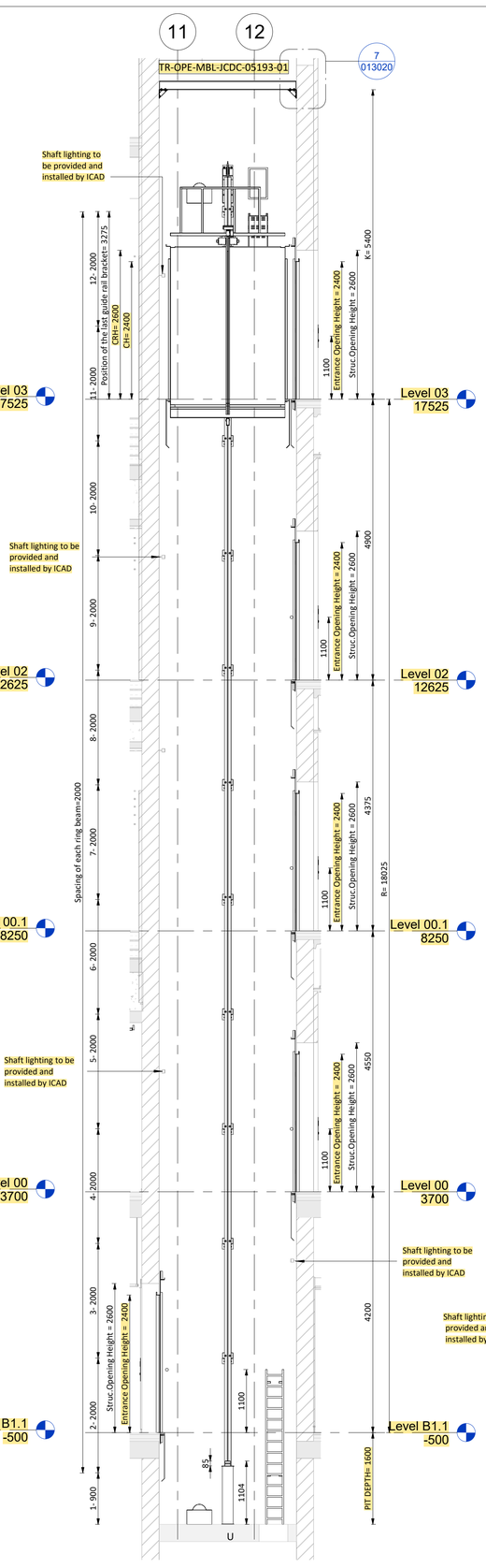
dar — Dar Al Handasah consultants (Shair and Partners)
Checked and Reviewed

LEGEND

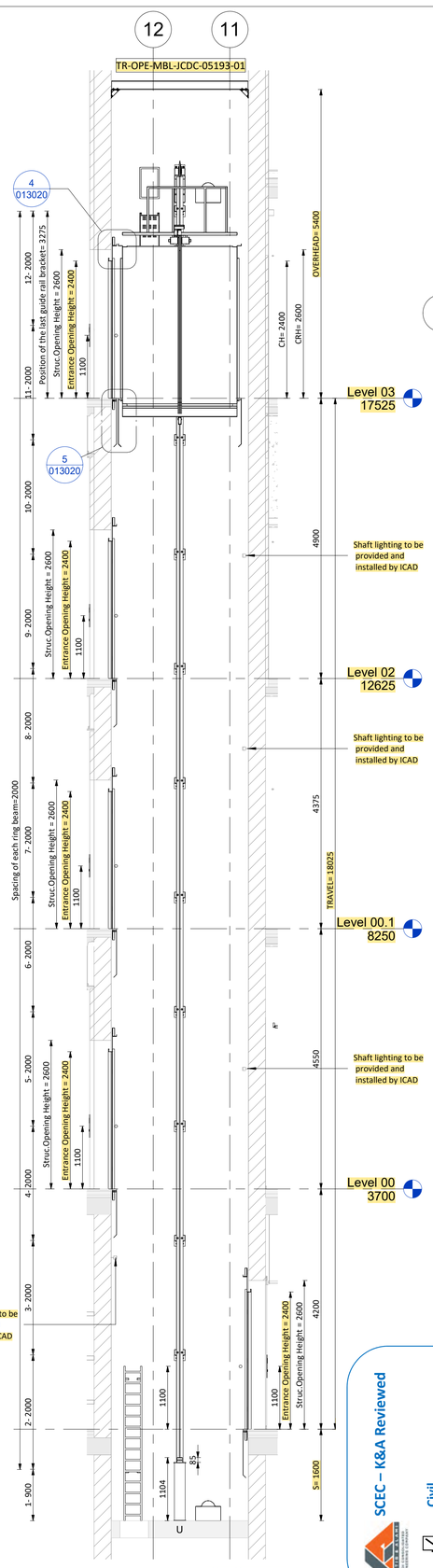
HW	HOIST WAY WIDTH
HD	HOIST WAY DEPTH
CRH	CAR ROUGH HEIGHT
DOOR OP	DOOR OPENING
ROP	ROUGH OPENING
CW	CAR WIDTH
CD	CAR DEPTH
CH	CAR HEIGHT
DBG	DISTANCE BETWEEN GUARDRAIL
CWT	COUNTERWEIGHT
E&I PANEL	EMERGENCY & INSPECTION PANEL
G.O.V	CAR GOVERNOR
COP	CAR OPERATING PANEL
HBP	HALL POSITION INDICATOR



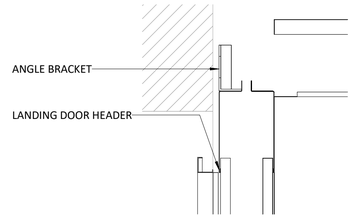
1 Section "A"-L21 & L22
1 : 50



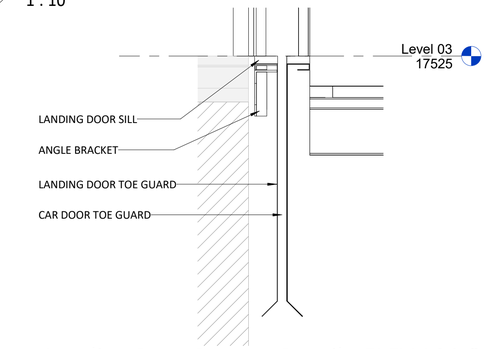
2 Section "B"-L21
1 : 50



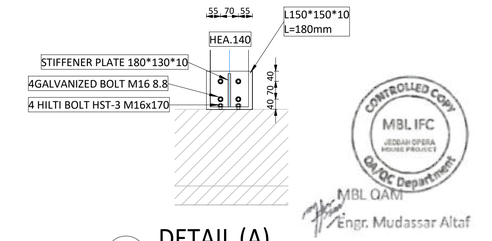
3 Section "C"-L22
1 : 50



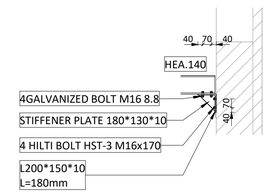
4 Landing Door Header Detail - ELEV -L21 & L22
1 : 10



5 Landing Door Bottom Detail - ELEV-L21 & 22
1 : 10



6 DETAIL (A)
1 : 12

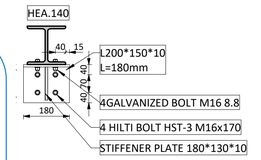


7 DETAIL (A1-A1)
1 : 25

SCEC – K&A Reviewed
Code B – Approved with Comments
(Do not Resubmit)

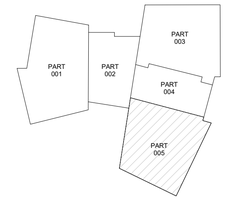
SCEC – K&A Reviewed

Civil	Arch.	Elec.	ELV.	Mech.	Acoustic
☒	☒	☒	☐	☒	☒



DETAIL (B1-B1)
1 : 12

dar Al-Handasah consultants (Shair and Partners)
Checked and Reviewed



KEY PLAN

GENERAL NOTES: ALL LEVELS SHOWN ON THE DRAWING ARE CHART DATUM LEVELS PROJECT LEVEL+0.000 = +3.700 CHART DATUM.

Refer to :
LIFTS overhead steel hook beams structural calculation and fixation details: TR-OPE-MBL-JCDC-05193-S01
CRS: SD-OPE-MBL-JCDC-01334_CRS- KA comment_1-version-(0_00)-L21,22
Electrical Panels distribution: JOH-ICD-S5-01-056-XXX-03-SD-EL-011630_1
wiring diagram opera lift
Acoustic Letter: L-OTIS-MBL-OPE-26-0015

DRAWING / DOCUMENT REFERENCE

NO.	NUMBER	TITLE	REV.
01	L-21 & L-22	JOH-HLA-S4-01-056-005-00-DR-AR-011105	F00
02		JOH-BHC-S4-01-056-XXX-XX-DR-VT-015001	
03		JOH-BHC-S4-01-056-XXX-XX-DR-VT-010001	
04		JOH-BHC-S4-01-056-005-ZZ-DR-VT-013001	
05		JOH-BHC-S4-01-056-005-ZZ-DR-VT-011001	
06		TQ-OPE-MBL-JCDC-00222	
07		JCD-OPE-MBL-MR-003-NMR-601-03	
08		JCD-OPE-MBL-MR-003-NMR-602-01	
09		TR-OPE-MBL-JCDC-05193-01	

REVIEWED BY DISCIPLINE	SIGNATURE	DATE
ARCHITECTURAL	MT	[illegible]
STRUCTURAL	PC	[illegible]
MECHANICAL	SbA	2026-05-20
PLUMBING	SbA	2026-05-20
ELECTRICAL	SbA	[illegible]
ICT	SbA	2026-05-[illegible]
ADDITIONAL DISCIPLINES TO BE ADDED AS REQUIRED		

REV	DATE	BY	DESCRIPTION	CHK	APD
S01	05-MAY-2026	AK	Shop Drawing	SS	AB
S00	05-MAY-2026	AK	Issued for Approval	SS	AB

DRAWING STATUS: CONSTRUCTION SHOPDRAWING

CLIENT: شركة وسط جدة للتطوير Jeddah Central Development Co.

LEAD DESIGN CONSULTANTS: Henning Larsen

LEAD ENGINEERING CONSULTANT: BURO HAPPOLD

PROJECT SUPERVISION CONSULTANT: KHATIB & ALAMI

GENERAL CONTRACTOR: MBL

REVIEWER: dar

SUB CONTRACTOR: OTIS ELEVATOR COMPANY SAUDI ARABIA LIMITED

PROJECT: JEDDAH OPERA HOUSE

TITLE: OPERA BUILDING
VERTICAL TRANSPORTATION
PART 005
SECTIONS - LIFT SHAFT (L21 & L22)

SCALE @ A1:	CHECKED:		APPROVED:
As indicated	SS		AB
PROJECT No:	DESIGNED:	DRAWN:	DATE:
440014	AK	PR	05-MAY-2026

DRAWING No: JOH-OTS-S5-01-056-005-ZZ-SD-VT-013020 REV: S01

MODEL FILE REF: JOH-OTS-S5-01-056-XXX-ZZ-M3-VT-010001

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